PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Supporting active ageing before retirement: a systematic review and meta-analysis of workplace physical activity interventions targeting older employees
AUTHORS	Merom, Dafna; Stanaway, Fiona; Gebel, Klaus; Sweeting, Joanna; Tiedemann, Anne; Mumu, Shirin; Ding, Ding

VERSION 1 – REVIEW

REVIEWER	Burn, Naomi Teesside University, Health and Life Sciences
REVIEW RETURNED	07-Dec-2020

GENERAL COMMENTS	Thank you for the opportunity to review this paper, which presents a systematic review and meta-analysis of workplace physical activity interventions for older adults. I would like to commend the authors for the preparation of such a well written manuscript. The paper presents a strong addition to the body of workplace PA literature. I do have some suggestions for the paper, which I have outlined in the form of major and minor comments below. Major Comments Introduction The introduction presents a strong rationale for conducting this review and is very clearly and succinctly written. I have presented a few minor comments for the introduction below. Methods I commend the authors for pre-registering their review protocol. However, the authors state that the protocol was first registered on 29th January 2018, whereas the manuscript states that the first search was conducted on 23rd December 2017. If this is correct, the protocol was not registered prior to the conduct of searches and therefore is it accurate to state that the protocol was registered a priori? Apologies if I have misunderstood. Page 8, line 82- Given critiques around the use of I squared in meta-analyses, I wondered whether the authors had considered using tau (or tau squared) rather than I squared to explore heterogeneity? I have provided some references below which may be useful. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2648991/ doi: 10.1186/1471-2288-8-79 https://handbook-5-1.cochrane.org/chapter_9/9_5_4_incorporating_heterogeneity_into_random_effects_models.htm

	I may have missed this, but did the authors consider any meta-regressions to explore the heterogeneity reported? Although this may not have been possible due to the small number of studies. Under the sub-groups section of the Prospero protocol the authors have stated that "Results will be stratified by type of intervention, outcome variables, quality of study, long and short term evaluation, reach, potential relevance to active ageing agenda". Was this conducted? Results According to the PRISMA diagram, 18 studies were included in the narrative review and 11 in the meta-analyses. Either in the text or the PRISMA diagram it would be useful to explain the reasons why those 7 studies were not included in the meta-analyses. Given that the authors meta-analysed several different outcomes, it might be useful to include details on the number of studies included in each outcome in the PRISMA diagram. At the moment the PRISMA diagram reads as if 11 studies were meta-analysed together, which is not the case. I realise this information is presented in another table, but it would be useful in the PRISMA diagram, for completeness. Page 18, line 20: The authors state that some studies report intervention fidelity measures. I may have missed it, but is this information presented anywhere in the tables or text? If not it would be useful to include this information in the paper. Discussion I would like to commend the authors for their considered recommendations presented in Table 4. There are several important points within this table. I think that the discussion would be strengthened drastically by exploring each of these points. As it stands the discussion does cover some of the points in Table 4, but some points are not expanded upon, and they are very important. I have outlined a few areas below that could be expanded upon in the discussion: • Table 4, number 2: The authors mention that of the studies included in their review, the workplaces were mostly university and hospital settings. In the discussion, could the authors comment on this, and perhaps propose why this might be? E.g. for example could it be the ease of access for academics to university workplaces? Also could the authors comment on why this could be an issue with the generalisability of the evidence base to other working environments? Are universities and hospitals representative of other working environments? • Table 4, number 6: the authors recommend better reporting of PA trials. Could the authors expand on this in the discussion, and perhaps give recommendations for how the reporting of PA trials could be improved (e.g. reporting checklists like Tidier Checklist?). • Table 4, number 7: The same point as above applies here regarding the poor reporting of intervention implementation fidelity (bullet point 7, Table 4). • Table 4, bullet point 9- should only quasi-experimental studies adjust for confounding variables? • Table 4- bullet point 10- what does harmonising measures across studies mean? Would standardise be a better word here? I may have missed it, but in the limitations section of the discussion did the authors mention the small number of studies included in each of the meta-analyses? Minor comments Abstract: I am not entirely sure the first line of the abstract makes sense. How would the health and fitness of older employees support Active Aging Policies? Perhaps this sentence would make more sense if this phrase were reversed. E.g. Active Aging

	Policies support the health and fitness of older employees. Although I may be interpreting the meaning incorrectly here. Page 4- Should the abbreviation for WHO be written out in full the first time it is used? Page 4 line 54- Here, I suggest the authors consider changing the wording of the phrase “good PA habits”, perhaps “maintaining adequate PA levels” or “adhering to PA guidelines” would be better? The phrase “good PA habits” is quite vague given the authors do not define what “good PA habits” are for this group. In line with the above comment, as one of the objectives of the review is to “assess the effect of interventions on PA behaviour and fitness outcomes highlighted in PA guidelines for older adults” it would be useful to include a sentence or two in the introduction specifying the physical activity guidelines for older adults. I notice this information is presented in the discussion, but it would also be useful earlier on in the paper. Page 7, line 17- was there a rationale for the age limit of ≥50 years of age? If so, could this be included in the inclusion/ exclusion criteria section? Page 7, line 24- How did the authors make the judgement that PA was a key component of an intervention? Was there a cut off for this decision? Page 7 line 36- the authors mention “research aim 1” here, but research objectives are stated in the introduction. Please ensure consistency of terms throughout the manuscript. Page 8 line 3- I realise the authors have stated which tool was used to assess risk of bias in the section above this, but given this section is named “quality assessments and analysis” it would be useful to re-state specifically which tools were used. Or perhaps the authors would prefer to move the information about study quality assessment tools from the section above to this section? Page 16, line 15- could the authors state here the number of studies that included strength/ balance activities? “Rarely” is quite an ambiguous term and requires the reader to search through the table for information. Given that this is one of the main conclusions presented in the abstract it would be useful to make this finding as clear as possible. Page 18, line 67- does targeted strategy need to be abbreviated here? Given that is it so infrequently used, I don’t think the abbreviation is required. Page 26, line 14: a reference is needed here Page 26, line 28- is “muscles strength” the correct term here? Appendix Table 1: Row containing information regarding references 20 and 21. Please could the authors spell out vegetables, rather than “vegie”. Would it be possible to include a key for Appendix Table 1? Some abbreviations like TTM and SCT and not written out in full in the table. In Table 4- could the authors expand on what they mean by “multi-dimensional” interventions? Do they mean interventions that combine aerobic and strength training PA prescriptions? Table 4, bullet point 5- typo Table 4, bullet point 12- typo

REVIEWER	Coca Nuñez, Aitor University of Deusto, Physical Activity and Sports Sciences
REVIEW RETURNED	07-Dec-2020

GENERAL COMMENTS	The paper is very relevant and well designed. It is crucial to realize that with an older workforce, PA in the workplace or PA interventions that target all WHO recommendations (balance, strength, moderate PA time) could be the key to maintain healthy lifestyles and quality of life. This review points out the lack of such evidences and the need for future interventions that target strength and balance as important qualities for this population. There are some minor revisions needed: Page 8. Line 22 medium (> 0.20-0.5) authors should review this as the have the limit for large effect as large (≥0.8) Page 18 Line 10 , and three interventions lasted up to 13 months [24, 32, 35] On table 1, only reference 32 last 13 months, the other two lasted 12 months, double check and correct Appendix Table 1, some of the references numbers are incorrect, check and improve. e.g. Bassey, E. J.,et al, 1987 [27] should be reference[30]

REVIEWER	Hurst , Christopher Newcastle University, Institute of Neuroscience
REVIEW RETURNED	13-Dec-2020

GENERAL COMMENTS	Thank you to authors for giving me the opportunity to review this interesting manuscript. Please see below for some general comments. General comments • The manuscript is generally very well written and is an enjoyable experience for the reader. Congratulations to the authors for this. • Clearer justification is needed as to why 50 years of age has been used as the cut off for 'older adults'. • I found the use of the term 'multiple risk factor interventions' confusing. Does this refer to interventions targeting multiple behaviours including PA? Also, I don't think this term is used within the method section. Please ensure consistency throughout the manuscript. • The article type on the submission currently states 'original research' rather than 'systematic review' Please see below for my specific comments on each section of the manuscript. Introduction • The introduction sets the scene well. I did wonder if the authors were able to acknowledge the positive effects of older adults in the workforce as the first two paragraphs are quite negative. Methods

	• Please provide full search strategy for at least one database as per Point 8 of PRISMA (I'm not clear why this has been marked NA on the PRISMA checklist submitted by the authors). This can be provided in a supplementary file rather than the main manuscript. • Section 2.1: Could the authors explicitly state what type of PA interventions they are interested it – from a delivery perspective. It is interventions which specifically prescribe exercise / PA? Interventions which consist of PA counselling? Or could it be either. A clear statement on this would be helpful to the reader. • Section 2.2: The authors refer to 'Supp 1'. I can't find anything in the supplementary information that seems relevant here. Apologies if I am missing something. • Section 2.2: Which version of endnote was used. • Quality assessment and analysis (line 13, page 8): should this read 'heterogeneity' rather than 'homogeneity'? Results • The results section is very long. There is a lot of information presented within the text in the results section of the manuscript. This made it quite a challenging read for me and I fear this may be the case for potential readers. There is considerable duplication in the information in the text and the tables. I suggest that the text is shortened to make this section more readable. • The SMD value for balance is different in Table 3 (1.02) compared with that on the Forest plots (1.08) Discussion • Generally, well written and puts the results into context. • I am not clear what Recommendation 1 (Table 4) means. Could the authors reword this for clarity. • Recommendation 10 (Table 4): I'm not sure if 'expected' is the most appropriate wording here. Perhaps 'specific' to these activities would be more appropriate. Tables and figures • • Please add reference numbers to the remainder of the studies included in 'Appendix Table 1' • Please add reference numbers to 'Supplementary Table 2' • Please add reference numbers to forest plots

REVIEWER	Thrift, Amanda Monash University, Department of Medicine, School of Clinical Sciences
REVIEW RETURNED	18-Jan-2021

GENERAL COMMENTS	This is a systematic review of physical activity interventions conducted in the workplace. Overall, the study appears well-conducted and the statistics well-described, although some additional analyses and details would enhance the value of the manuscript, as outlined below. 1. The authors should include the search strategy used to identify all articles. This is a PRISMA criterion and will allow others to repeat the search.

	2. Figure 1 shows that 7 studies were included in a qualitative synthesis only, while 11 were included in both a meta-analysis and qualitative analysis. In Table 2, the bottom row indicates 14 studies with quantitative outcomes, so this is confusing. It would be helpful for readers to be able to identify which studies were included in each. Presumably, there were too few studies to conduct a meta-analysis on some outcomes. Please clarify in the table. 3. The supplementary figure would benefit from further details, including a title, the number of participants in each group with positive/negative outcomes, the study weight (i.e. its contribution to the effect size), and arrows at the bottom of each graph to show the direction of effect, i.e. intervention better or intervention worse. A short description of how weighting was calculated should be included in the methods (presumably this was done as the point estimates in the supplementary figure differ in size). 4. Some explanation of the GRADE summary should be included in the legend of Table 3 (and Supplementary Table 2). It would also be helpful for readers if the authors included a graph that summarized the risk of bias in the main paper. The importance of the take-home message in a visual display should not be underestimated.

VERSION 1 – AUTHOR RESPONSE

Response to Reviewer 1

Comments	Responses	Page
Abstract: I am not entirely sure the first line of the abstract makes sense. How would the health and fitness of older employees support Active Aging Policies? Perhaps this sentence would make more sense if this phrase were reversed. E.g. Active Aging Policies support the health and fitness of older employees. Although I may be interpreting the meaning incorrectly here	Thanks for your comment. We begged to differ. The active ageing policies recommend keeping older adults engaged with paid employment as long as possible. For this to be possible the physical and cognitive capacities must be preserved when they are still employed, which can be addressed by regular exercise. So the logic order is that exercise can support this new policy.	
Introduction: The introduction presents a strong rationale for conducting this review and is very clearly and succinctly written. I have presented a few minor comments for the introduction below	We thank the reviewer for acknowledging the clarity of the introduction.	
Minor comments to introduction.		
Page 4- Should the abbreviation for WHO be written out in full the first time it is used?	We have now used World Health Organisation (WHO) as it is the first time mentioned.	Page 4

Page 4 line 54- Here, I suggest the authors consider changing the wording of the phrase “good PA habits”, perhaps “maintaining adequate PA levels” or “adhering to PA guidelines” would be better? The phrase “good PA habits” is quite vague given the authors do not define what “good PA habits” are for this group	We agree. We have now changed it as follow: “Adherence to older adults’ comprehensive PA guidelines including balance exercises, muscle strengthening and moderate-intensity aerobic activity, is likely to benefit older employees in terms of health as well as work productivity and safety.	Page 4
In line with the above comment, as one of the objectives of the review is to “assess the effect of interventions on PA behaviour and fitness outcomes highlighted in PA guidelines for older adults” it would be useful to include a sentence or two in the introduction specifying the physical activity guidelines for older adults	We agree. We have now introduced the broader guidelines in short to keep the fluency of the sentence (balance exercises, muscle strengthening and moderate-intensity aerobic activity). In the discussion we also specify the recommended frequency and duration for each type of PA in the guideline.	Page 4
Methods – Major comments		
I commend the authors for pre-registering their review protocol. However, the authors state that the protocol was first registered on 29th January 2018, whereas the manuscript states that the first search was conducted on 23rd December 2017. If this is correct, the protocol was not registered prior to the conduct of searches and therefore is it accurate to state that the protocol was registered a priori? Apologies if I have misunderstood.	We submitted the protocol on the 21 of December 2017. We received a message on the site that due to the holiday season expect delays. Only on the 22/01/2018 we received email acknowledging our submission and eight days later, on the 29/01 (1.59AM of the 30/01 Sydney time) we received the registration number. We insert this information but since we started our database search before official registration, we omitted the ‘a-priori’ from the dot points on strengths and limitations.	Page 3
Page 8, line 82- Given critiques around the use of I squared in meta-analyses, I wondered whether the authors had considered using tau (or tau squared) rather than I squared to explore heterogeneity? I have provided some references below which may be useful	I thank the reviewer for this suggestion. Each measure gives a different perspective of the results rather than being more appropriate. In fact Tau is reflected by the 95% confidence interval of the ES . Therefore we reported on the Tau-square which indicates how widely the effects vary in various population and the I^2 measure which indicates the proportion attributed to between-studies variability.	Page 9 Table 3
I may have missed this, but did the authors consider any meta-regressions to explore the heterogeneity reported?	We were hoping to explore sources of heterogeneity, but the number of trials were too small. This is one of the	

Although this may not have been possible due to the small number of studies.	findings that there were not many interventions targeting older employees, and few reported on similar outcome.	
Under the sub-groups section of the Prospero protocol the authors have stated that “Results will be stratified by type of intervention, outcome variables, quality of study, long and short -term evaluation, reach, potential relevance to active ageing agenda”. Was this conducted?	We provided only narrative summary as due to the small number of trials sub- group analyses was not possible. Table 2 in the main manuscript summarises all the intervention characteristics based on Supp Table 1. Supp Table 2 presents the quality of each study but again, sub-analysis by study quality couldn’t be done due to the small number of studies.	
Methods- minor comments		
Page 7, line 17- was there a rationale for the age limit of ≥50 years of age? If so, could this be included in the inclusion/ exclusion criteria section?	The definition of “older worker” is controversial and ranges from 40 to 65. For example, in the US Labour department it was defined as 55 years of age whereas in the workforce sector a person aged 45 and over is generally consider an “older” employee. Our decision was data driven from the previous SR by Posica et al. (2016) who systematically searched for health promotion interventions that explicitly targeted “older employees” regardless of age. One of their findings was that participants mean age ranged from 51- to 55 years. Our decision was therefore based on that minimum considering it was found to be typically represent most participants in worksite health promotion interventions for older adults. We have now added short explanation reflecting this decision.	Page 6
Page 7, line 24- How did the authors make the judgement that PA was a key component of an intervention? Was there a cut off for this decision?	Thank you for this comment. We now omitted the word “key” because it does not reflect what we did. We included multiple risk behaviour interventions as long as PA was addressed in the intervention and there was an assessment of any PA outcome.	Page 6
Page 7 line 36- the authors mention “research aim 1” here, but research objectives are stated in the introduction. Please ensure consistency of terms throughout the manuscript.	Thank you. We now removed the statement “research aim 1”	Page 7

Page 8 line 3- I realise the authors have stated which tool was used to assess risk of bias in the section above this, but given this section is named “quality assessments and analysis” it would be useful to re-state specifically which tools were used. Or perhaps the authors would prefer to move the information about study quality assessment tools from the section above to this section?	We agree. We moved the specification of tools used to the section “Quality assessment and analysis” as suggested. Thank you.	Page 8
Results – Major comments		
According to the PRISMA diagram, 18 studies were included in the narrative review and 11 in the meta-analyses. Either in the text or the PRISMA diagram it would be useful to explain the reasons why those 7 studies were not included in the meta-analyses. Given that the authors meta-analysed several different outcomes, it might be useful to include details on the number of studies included in each outcome in the PRISMA diagram. At the moment the PRISMA diagram reads as if 11 studies were meta-analysed together, which is not the case. I realise this information is presented in another table, but it would be useful in the PRISMA diagram, for completeness.	We now included 12 studies in meta-analysis and we added to the PRISMA diagram that a maximum of 3 studies were meta-analysed per outcome, as the abstract previously specified. We provided the reasons for exclusion of two studies in the PRISMA as well as in the results section on Trials effectiveness.	Page 21
Page 18, line 20: The authors state that some studies report intervention fidelity measures. I may have missed it, but is this information presented anywhere in the tables or text? If not it would be useful to include this information in the paper	In Table 2 -summary of study characteristics under Process data the letter ‘F’ stands for fidelity (explained in the legend). The exact information is presented in Supplementary Table 2 under the column “Process data” we extracted all the information on the process of implementation including the details given on fidelity which is now underline.	
Results- minor comments		
Page 16, line 15- could the authors state here the number of studies that included strength/ balance activities? “Rarely” is quite an ambiguous term and requires the reader to search through the table for information. Given that this is one of the main conclusions presented in the abstract it would be useful to make this finding as clear as possible.	Agree! We have now detailed the five studies that focused on muscle strength and/or balance and/or flexibility.	

Page 18, line 67- does targeted strategy need to be abbreviated here? Given that is it so infrequently used, I don't think the abbreviation is required	This abbreviation has been deleted. It appears in Table 2 legend.	
Discussion- major comments		
Table 4, number 2: The authors mention that of the studies included in their review, the workplaces were mostly university and hospital settings. In the discussion, could the authors comment on this, and perhaps propose why this might be? E.g. for example could it be the ease of access for academics to university workplaces? Also could the authors comment on why this could be an issue with the generalisability of the evidence base to other working environments? Are universities and hospitals representative of other working environments?	We agree with the reviewer that the setting choices can be influenced by the ease of conducting a research as well as environment that conducive to health messages. We have included short paragraph in the discussion that referred to the limited generalisability of the interventions as follow: *It is worth noting that about 40% of the interventions were conducted in universities or academic hospitals/clinics. This could possibly be due to ease of doing research in these settings coupled with environments that are conducive to health promotion messages, such as health-care providers or academics. Hence, the generalisability of intervention implementations and the findings of these studies is limited"*	Page 27
Table 4, number 6: the authors recommend better reporting of PA trials. Could the authors expand on this in the discussion, and perhaps give recommendations for how the reporting of PA trials could be improved (e.g. reporting checklists like Tidier Checklist?).	We agree; we have now included a new paragraph stating the limited reporting of PA component in multiple risk factors intervention which limit their replication. We referred to TIDieR-as methods of improvement. *Multiple risk factor interventions mostly had poor description of the PA component of the intervention which reduced their reproducibility. For example, referrals to exercise lacked specifications of the type of exercise programs that were available to clients, or goal settings lacked descriptions of how goals were tailored to individuals, how they were monitored and by whom. In addition, most studies rarely reported the fidelity of the interventions. Hence it is recommended to use a template for reporting of PA interventions, such as TIDieR [51], in order to allow better replication of sensible effective PA interventions.*	Page 27

Table 4, number 7: The same point as above applies here regarding the poor reporting of intervention implementation fidelity (bullet point 7, Table 4).	This has been addressed in the same paragraph we added referring to TIDieR to allow replication of sensible effective PA interventions	Page 29
Table 4, bullet point 9- should only quasi-experimental studies adjust for confounding variables?	We agree. Confounding is not limited to quasi-experimental studies. Cluster RCTs certainly prone to it and sometime unsuccessful randomisation in RCTs. We now made our suggestion broader as follow: "*Studies should assess confounders and adjust for their presence*"	Table 4 methodology section
Table 4- bullet point 10- what does harmonising measures across studies mean? Would standardise be a better word here?	We agree, we replaced the word to 'standardised'.	
I may have missed it, but in the limitations section of the discussion did the authors mention the small number of studies included in each of the meta-analyses?	Thanks for alerting this issue. We mentioned this in the abstract but failed to include in the section of study limitations. We have now mentioned this as a limitation.	Page 29
Discussion- minor comments		
Page 26, line 14: a reference is needed here	We now have added the reference that exercise can prevent falls.	
Page 26, line 28- is "muscles strength" the correct term here?	Corrected	
Appendix Table 1: Row containing information regarding references 20 and 21. Please could the authors spell out vegetables, rather than "vegie".	Corrected to "vegetables"	
Would it be possible to include a key for Appendix Table 1? Some abbreviations like TTM and SCT and not written out in full in the table.	We have now spelled out TTM as Transtheoretical Model and SCT as Social Cognitive Theory along the Table.	
In Table 4- could the authors expand on what they mean by "multi-dimensional" interventions? Do they mean interventions that combine aerobic and strength training PA prescriptions?	Yes, we meant interventions that addresses more than one fitness dimension. The sentence was deleted as it doesn't add much to the preceded sentence.	Anne
Table 4, bullet point 5- typo	Thank you!. We correct this to frailty rather than fragility	
Table 4, bullet point 12- typo	Thank you. Corrected to Sustainability.	

Response to Reviewer 2

The paper is very relevant and well designed. It is crucial to realize that with an older workforce, PA in the workplace or PA interventions that target all WHO recommendations (balance, strength, moderate PA time) could be the key to maintain healthy lifestyles and quality of life. This review points out the lack of such evidences and the need for future interventions that target strength and balance as important qualities for this population.	We thank the reviewer for acknowledging the relevance and importance of this Systematic review	
Page 8. Line 22 -medium (> 0.20-0.5) authors should review this as the have the limit for large effect as large (≥0.8)	This has now been corrected covering a full range…" *Medium (> 0.20-0.79) or large (≥0.8)"*	Page 8
Page 18 Line 10 , and three interventions lasted up to 13 months [24, 32, 35] On table 1, only reference 32 last 13 months, the other two lasted 12 months, double check and correct	The Table is correct; we have now changed the text as follow: *"three interventions lasted 12 to 13 months"*	Page 18
Appendix Table 1, some of the references numbers are incorrect, check and improve. e.g. Bassey, E. J.,et al, 1987 [27] should be reference[30]	Thank you, we have now fixed the references in the Supplementary Table 1.	

Responses to Reviewer 3

The manuscript is generally very well written and is an enjoyable experience for the reader. Congratulations to the authors for this.	Thank you.	
Clearer justification is needed as to why 50 years of age has been used as the cut off for 'older adults'.	This comment was raised by reviewer 1 as well and we now added the justification.	Page 6
I found the use of the term 'multiple risk factor interventions' confusing. Does this refer to interventions targeting multiple behaviours including PA? Also, I don't think this term is used within the method section. Please ensure consistency throughout the manuscript.	Thank you for alerting us to this issue. We now added this clarification in our methods section as follow: "*Studies that targeted other health risks in addition to physical inactivity (hereafter, multiple risk factors interventions) were only included if….."*	Page 6

The article type on the submission currently states 'original research' rather than 'systematic review'	The option Systematic Review was not available in the site	
Introduction		
The introduction sets the scene well. I did wonder if the authors were able to acknowledge the positive effects of older adults in the workforce as the first two paragraphs are quite negative	Thank you. We agree. We now have highlighted the positive aspect of ageing as follow: "*While a number of mental and social abilities can increase with age, and such changes should be viewed as strengths and be respected [4], age-related physiological decline…*	Page 4
Methods		
Please provide full search strategy for at least one database as per Point 8 of PRISMA (I'm not clear why this has been marked NA on the PRISMA checklist submitted by the authors). This can be provided in a supplementary file rather than the main manuscript	We have now provided our key term mapping and example of Medline search as supplementary file	
Section 2.1: Could the authors explicitly state what type of PA interventions they are interested it – from a delivery perspective. It is interventions which specifically prescribe exercise / PA? Interventions which consist of PA counselling? Or could it be either. A clear statement on this would be helpful to the reader	PA is a broad concept by definition; it will include any movement that involved large body muscles. We gave examples of such interventions. (Page 6, lines 19-24). To avoid confusion we have included in the first sentence the following: *"We included studies that involved any PA interventions (i.e, intervention that will encourage people to move large skeletal muscles)*	Page 6
Section 2.2: The authors refer to 'Supp 1'. I can't find anything in the supplementary information that seems relevant here. Apologies if I am missing something.	We apologise. It is Supplementary file, which should have been added to the submission detailing the search strategy as submitted to PROSPERO. We have now added it with exemplar of on database search.	
Section 2.2: Which version of endnote was used.	We added now the version number which was X9.3.3.	
Quality assessment and analysis (line 13, page 8): should this read 'heterogeneity' rather than 'homogeneity'?	This was a typo. We have now corrected this.	
Results		

The results section is very long. There is a lot of information presented within the text in the results section of the manuscript. This made it quite a challenging read for me and I fear this may be the case for potential readers. There is considerable duplication in the information in the text and the tables. I suggest that the text is shortened to make this section more readable	The paper had three objectives that is what makes it long. We therefore used subheadings to ease the readability. We tried our best to reduce words but we could only cull 170 words. We hope this is satisfactory and we were within the SR word limit.	
The SMD value for balance is different in Table 3 (1.02) compared with that on the Forest plots (1.08)	Thank you. We repeated our analyses to provide tau-square (reviewer 1 asked for it) and we found that our balance estimate was lower and we changed it to 0.74.	
Discursion		
Generally, well written and puts the results into context.	Thank you	
I am not clear what Recommendation 1 (Table 4) means. Could the authors reword this for clarity.	We have now reword this point as follow: "*To enable calculation of the reach of workplace interventions we suggest that as a first step employers will endorse the intervention and provide the list of all employees or at least the number of the potential target population (e.g., age or year prior to retirement). It may also boost recruitment when the management of the workplace endorses such intervention*"	Page 29 Table 4 Reach and recruitment
Recommendation 10 (Table 4): I'm not sure if 'expected' is the most appropriate wording here. Perhaps 'specific' to these activities would be more appropriate	We change it to 'specific'	
Please add reference numbers to the remainder of the studies included in 'Appendix Table 1'	Done	
Please add reference numbers to 'Supplementary Table 2'	Done	
Please add reference numbers to forest plots	References are provided in Table 3 that presents the effect sizes . The forest plots contains details such as trial weight author and year etc. Note that when we conducted forest plot the final references as in the manuscript was not known.	

Responses to Reviewer 4

The authors should include the search strategy used to identify all articles. This is a PRISMA criterion and will allow others to repeat the search	We now provided it in the Supplementary file	
Figure 1 shows that 7 studies were included in a qualitative synthesis only, while 11 were included in both a meta-analysis and qualitative analysis. In Table 2, the bottom row indicates 14 studies with quantitative outcomes, so this is confusing. It would be helpful for readers to be able to identify which studies were included in each. Presumably, there were too few studies to conduct a meta-analysis on some outcomes. Please clarify in the table.	We now explain the exclusions of 6 studies from meta-analysis in the PRISMA flow chart. 14 individual studies had quantitative outcomes, but two studies cannot be used due to results not allowing extraction of information as authors used mixed model with interaction term.	Figure 1 Page 21
The supplementary figure would benefit from further details, including a title, the number of participants in each group with positive/negative outcomes, the study weight (i.e. its contribution to the effect size), and arrows at the bottom of each graph to show the direction of effect, i.e. intervention better or intervention worse. A short description of how weighting was calculated should be included in the methods (presumably this was done as the point estimates in the supplementary figure differ in size).	We re-produce the figures to include the study weight and the direction of the effect (favour exercise or favour control). Titles were also in the previous forest plots and included in the new figures. The space in the forest plot is too limited to add also the n, which is anyway reported in Table 1 and 3.	
Some explanation of the GRADE summary should be included in the legend of Table 3 (and Supplementary Table 2). It would also be helpful for readers if the authors included a graph that summarized the risk of bias in the main paper. The importance of the take-home message in a visual display should not be underestimated.	We have now included in the legend of Table 3 explanation of GRADE levels. The Supplementary Table 2 refers to ROB; assessment of each domain includes too many details to be added as legend. We now produce the figure (Supp Fig1)	

VERSION 2 – REVIEW

REVIEWER	Burn, Naomi Teesside University, Health and Life Sciences
REVIEW RETURNED	18-Feb-2021

GENERAL COMMENTS	The authors have attended to my comments well and I have no further comments. However, I cannot find the updated PRISMA flow chart in the submitted files or manuscript, although I do not

	feel the need to see the manuscript again when this has been amended.

REVIEWER	Hurst , Christopher Newcastle University, Institute of Neuroscience
REVIEW RETURNED	09-Mar-2021

GENERAL COMMENTS	No further comments.

REVIEWER	Thrift, Amanda Monash University, Department of Medicine, School of Clinical Sciences
REVIEW RETURNED	25-Feb-2021

GENERAL COMMENTS	Statistical review: This is a systematic review of physical activity interventions conducted in the workplace. The authors have addressed some of my concerns. A couple of outstanding issues remain. 1. While the authors have amended their PRISMA flow diagram, it remains unclear which studies were included in the quantitative analysis. I suggest that the authors either include another row in Table 2 that shows which studies were included in the quantitative component. Alternatively, this could be described in a footnote to this table. 2. While the authors do have a title for Supplementary Figure 1, there is no detail about what each of the levels of mean (i.e. low, medium, and high). Each figure should be able to be understood without reference to the text (or to Tables 1 and 2). Please include the appropriate details in the footnote to the figure so that it can be understood on its own. Similarly, Supplementary Figure 2 is missing details about the outcome, or any other context that would enable it to be understood without reference to the text. This figure is also missing a title. The authors state that titles are provided, but the document that I have to review does not. The authors similarly state that the space in the forest plot is too limited to include the n's. However, these graphs appear to be created in stata; this program does have the capability to include more columns of test. Please amend. 3. The authors state that there are too many details for to add about the assessment of risk of bias to Supplementary Table 2. At the very least the authors should include a reference in the footnote, so that readers can readily find these details if they wish.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1- Dr. Naomi Burn, Teesside University	Responses
The authors have attended to my comments well and I have no further comments. However, I cannot find the updated PRISMA flow chart in the submitted files or manuscript, although I do not feel the need to see the manuscript again when this has been amended.	*Thank you for acknowledging that we have addressed all your comments. The Figure was uploaded separately as JPG file*

Reviewer: 4 -Prof. Amanda Thrift, Monash University	
While the authors have amended their PRISMA flow diagram, it remains unclear which studies were included in the quantitative analysis. I suggest that the authors either include another row in Table 2 that shows which studies were included in the quantitative component. Alternatively, this could be described in a footnote to this table.	*Thank you for alerting us to the issue. We have now included in Table 2 a line which indicated the studies that were included and not included in the quantitative analysis. We found that 11 RCTs were included (as per our first submission); in two RCT data could not be extracted and in another RCT (Bassey et al 1983), the changes in walking behaviour was given* ***for intervention and control combined for*** *1/2 of the commencing sample. We therefore amended the Flow-Chart indicating 11 studies included in quantitative evaluation, Table 2 indicates studies included and in the Results section* ***3.5: effectiveness of intervention*** *we added to the text the following: "*Another RCT assessed changes in daily walking on half of the sample but reported the pre-post changes for the intervention and control groups combined and therefore was not included in meta-analysis [30].
2. While the authors do have a title for Supplementary Figure 1, there is no detail about what each of the levels of mean (i.e. low, medium, and high). Each figure should be able to be understood without reference to the text (or to Tables 1 and 2). Please include the appropriate details in the footnote to the figure so that it can be understood on its own.	The legend of Supplementary Figure 1 was always there – we now expanded the text to denote High Risk of bias, Low Risk of bias and unclear risk exactly as the Cochrane's ROB presentation.
3. Similarly, Supplementary Figure 2 is missing details about the outcome, or any other context that would enable it to be understood without reference to the text. This figure is also missing a title. The authors state that titles are provided, but the document that I have to review does not. The authors similarly state that the space in the forest plot is too limited to include the n's. However, these graphs appear to be created in stata; this program does have the capability to include more columns of test. Please amend.	Supplementary Figure 2 – is a that included all the forest plots. Each outcome is specified at the top of each figure. We change the title of the file to mention all the outcomes "Supplementary figure 2_Intervention effects on physical activity and fitness outcomes" We now included in each figure the n's for intervention and control.
The authors state that there are too many details for to add about the assessment of risk of bias to Supplementary Table 2. At the very least the authors should include a reference in the footnote, so that readers can readily find these details if they wish.	We added the references to the details of Supplementary Table 2 both for ROB for the RCTs as well as reference to the tool used for non-randomised interventions ROBINS-I